



Grade:

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29

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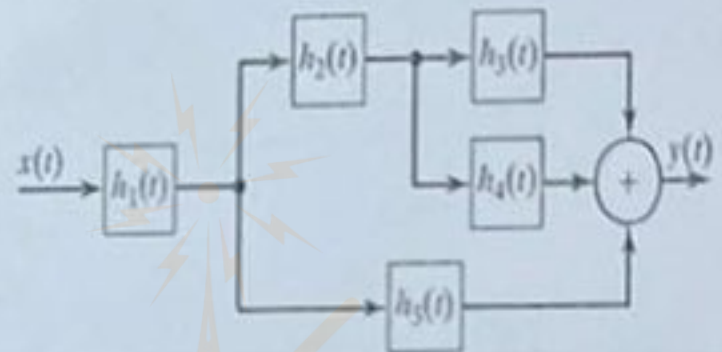
Q1. Select the correct answer for the following questions:

1. Which of the following is not true?

- a. $t^2 \delta(t) = 0$ b. $\cos(t) \delta(t) = \delta(t)$ ☒ c. $\sin(t) \delta(t - \pi) = \delta(t - \pi)$ d. $\int_{-\infty}^{\infty} (2t^2 + 1) \delta(t) dt = 1$

2. The total impulse response of the LTI system given below is.....

- ☒ a. $h(t) = h_1(t) * ([h_2(t) * (h_3(t) + h_4(t))] + h_5(t))$
b. $h(t) = h_5(t) * ([h_2(t) * (h_3(t) + h_4(t))] + h_1(t))$
☒ c. $h(t) = h_1(t) + ([h_2(t) * (h_3(t) * h_4(t))] + h_5(t))$
d. $h(t) = h_2(t) + ([h_1(t) * (h_3(t) + h_4(t))] + h_5(t))$



3. A system is given as $y(t) = \cos(x(t)) + 2 \sin(x(t-1))$. This system is described as

- ☒ a. Causal and memory. b. Stable and memory. c. Stable and memoryless. d. Non-causal and memory.

4. If $x(t) = e^{\cos(t)}$, then the signal is.....

- a. Odd and periodic signal. ☒ b. Even and periodic signal. c. Odd signal. ☒ d. Neither even nor odd.

5. The result of $\int_{-2}^6 5 \delta(t+4) dt$ is given as.....

- ☒ a. 0 b. 5t c. 5t + 20 d. 5

6. The power of the signal $s(t) = 8 \cos(20\pi t - \frac{\pi}{2}) + 4 \sin(15\pi t)$ is.....

- a. 12 b. 82 ☒ c. 40 d. 64

7. The function $x(t) = 6 \text{rec}(t/30)$ has a value of.....

- a. 1 for $-30 \leq t \leq 30$.
b. 6 for $-30 \leq t \leq 30$.
c. 1 for $-15 \leq t \leq 15$.
☒ d. 6 for $-15 \leq t \leq 15$.
e. 1 for $-60 \leq t \leq 60$.
f. 3 for $-60 \leq t \leq 60$.

8. Non-periodic signals are the examples of signals with

- ☒ a. Energy, Zero average power. ☒ b. Power, Zero energy. c. Energy, ∞ average power. d. Power, ∞ energy.

9. The fundamental period of the signal $x(t) = \cos(6t/4 + 40^\circ) + 2 \sin(8t/3)$ is:
 a. $\pi/4$ b. 12π c. $3\pi/4$ d. $\pi/3$

10. The result of $\cos(2t) + \cos(-2t)$ is

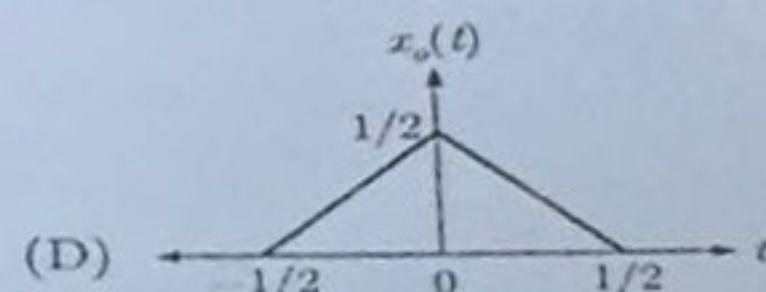
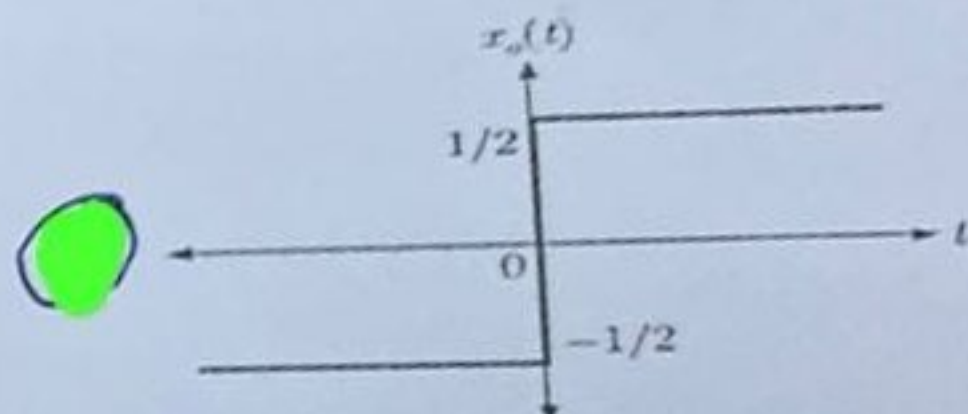
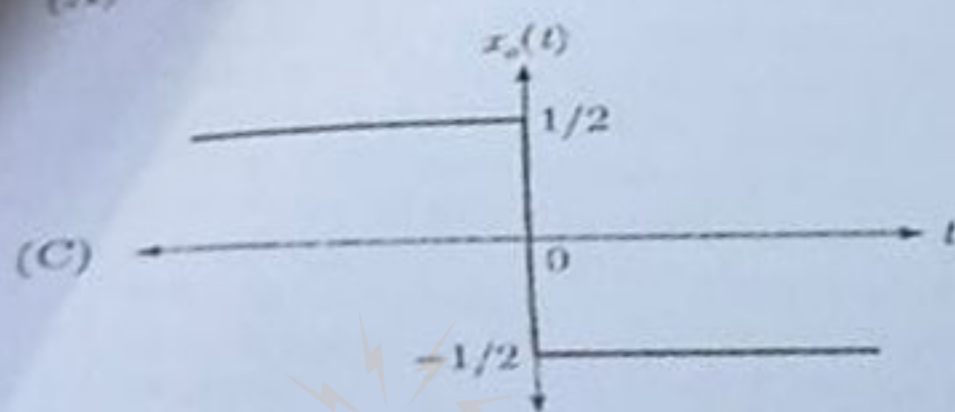
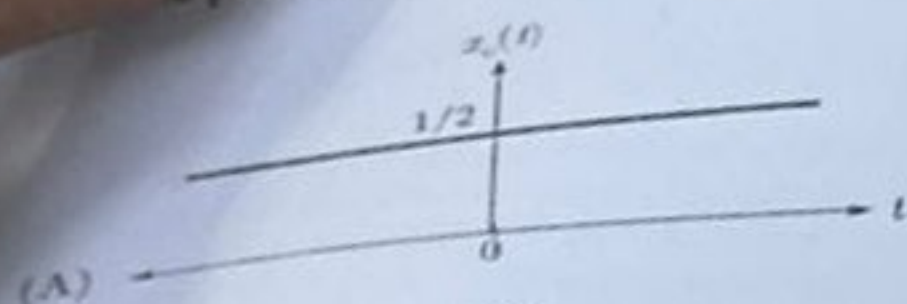
- a. $0.5 \cos(t)$ b. $e. 2 \cos(t)$ c. $2 \cos(4t)$ d. $0.5 \cos(2t)$ **$2 \cos(2t)$**

11. Which of the following system is causal?

- a. $y[n] = 2x[n] - 3x[n+1]$
 b. $y[n] = 2x[n]$
 c. $y[n] = 2x[n] - 3x[n+1]$
 d. $y[n] = 2x[n+1]$

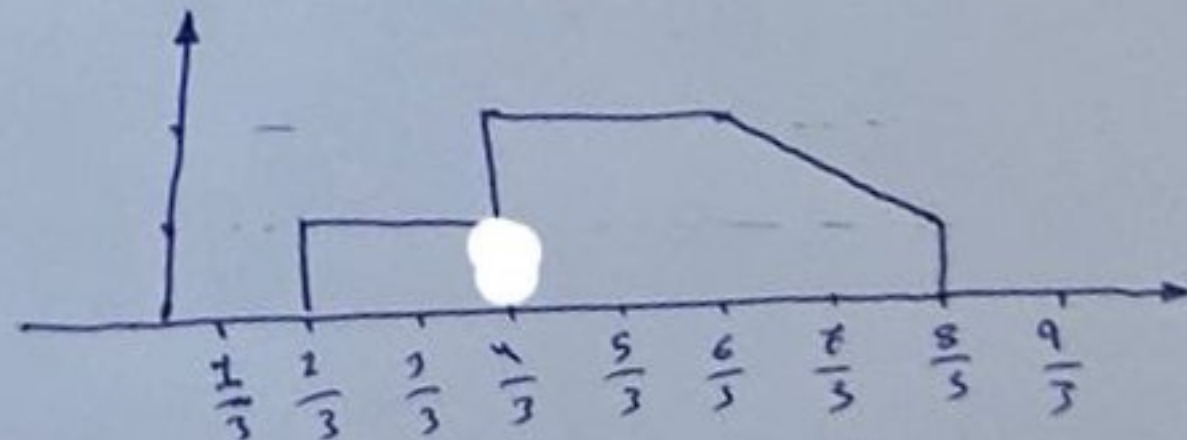
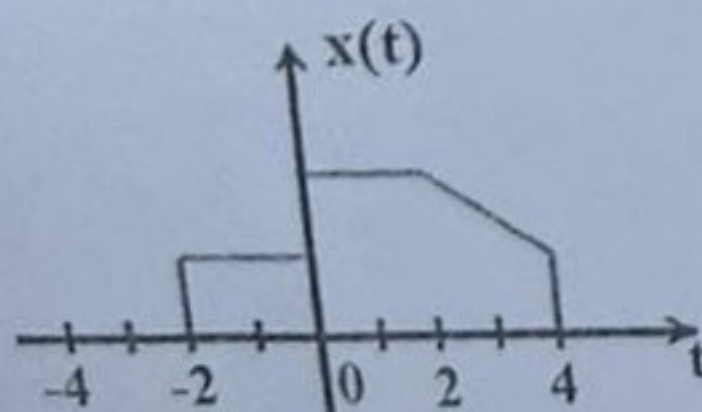
system is a system.
 a. Time-invertible, causal. b. Stable, memoryless. c. Stable, memoryless. d. Memoryless, causal. **Causal, stable**

part of a unit step signal is.....



Q2. Answer the following:

2.1. For the signal $x(t)$ shown below, sketch $x(3t - 4)$.



2. Express the following figure in a unit step form as.....

$u(t-1) + u(t-2) + u(t-3) - 3u(t-4)$

